

Shadman Salif Swanan

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Professional Summary

Analytical Computer Science graduate with strong skills in Data Science and Machine Learning. Experienced in using Python, SQL, and Power BI for data extraction, transformation, and visualization. Proficient in SQA methodologies, test case design, and API testing using Postman, with a solid foundation in software development.

Technical Skills

Programming Languages: Python, C++, JavaScript, PHP

Data Analysis: MySQL, Power BI, Microsoft Excel (Advanced), Process Modeling, EDA

AI/ML Frameworks: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Tensorflow, Hugging Face

Web & Tools: React, Vite, Node.js, Express.js, MongoDB, Git & GitHub, Tailwind CSS, Figma

SQA & Testing: Postman (API Testing), Test Case Design

Education

BRAC University

Bachelor of Science in Computer Science and Engineering

2022 – 2026

CGPA: 3.83 / 4.00 (Merit Based Scholarship)

Notre Dame College

Higher Secondary Certificate (Science)

2018 – 2021

GPA: 5.00 / 5.00

Ideal School and College

Secondary School Certificate (Science)

2008 – 2018

GPA: 5.00 / 5.00 (General Scholarship/ Dhaka Board)

Research Experience

Undergraduate Thesis Research | BRAC University

2025 – 2026

- Conducting research on semantic drift in Bengali using transformer representations. Built OCR pipelines and fine-tuned Historical BanglaBERT on a 120-year corpus, demonstrating strong data processing skills.

Real-Time Fraud Detection System | Independent

May 2026

- Built a fraud detection pipeline using LightGBM. Performed extensive exploratory data analysis (EDA) and threshold tuning.

Projects

Shikkha.com | React, Vite, Node.js, Express.js, MongoDB, JWT, Stripe, Tailwind CSS

- Developed a full-stack EdTech platform supporting course enrollment, online exams with automated grading, educator appointments, and e-commerce for books. Integrated JWT-based authentication, role-based access control, and Stripe Checkout.

Predictive Telecom Churn Optimization Framework | Python, Scikit-learn, Pandas, EDA, Feature Engineering

- Engineered a machine learning pipeline processing 100K customer logs with an advanced Gradient Boosting model, achieving a 0.6900 ROC-AUC score and 64.00% churn recall to project an estimated 2.22M USD in annualized retention savings.

Token-Level POS Tagging Using RNN Models | Python, TensorFlow, RNN, LSTM, GRU, BiLSTM, NLP

- Implemented and compared RNN, LSTM, GRU, and BiLSTM architectures for token-level part-of-speech tagging, evaluating sequence labeling performance across multiple recurrent neural network models.

Federal Flight Agency Mission-Critical Aerospace Network Design | Cisco Packet Tracer, RIPv2, Static Routing, VLSM

- Designed a high-availability network for autonomous aircraft operations using VLSM and a hybrid routing strategy combining RIPv2 and static routes to ensure automatic failover and reliable communication.

Experience & Certifications

GCI World Program

Apr 2026 – Present

The University of Tokyo (Matsuo-Iwasawa Laboratory)

Remote

- Participating in an international AI education program focused on machine learning and data science.

UX Design Decision-Making with AI & Design System Thinking with AI — GP Academy, 2026

Conference Management Officer — Notre Dame College Model United Nations, 2019

Co-Curricular Activities

Vice President, Photography Dept, NDNSC | Champion, BRACU Intra-Spring 2022 Football Tournament.

Runners-Up in Crusade 2026, RS Football Tournament | Completed Stage Drama Course (BRACU RS).

References

Dr. Chowdhury Mofizur Rahman

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